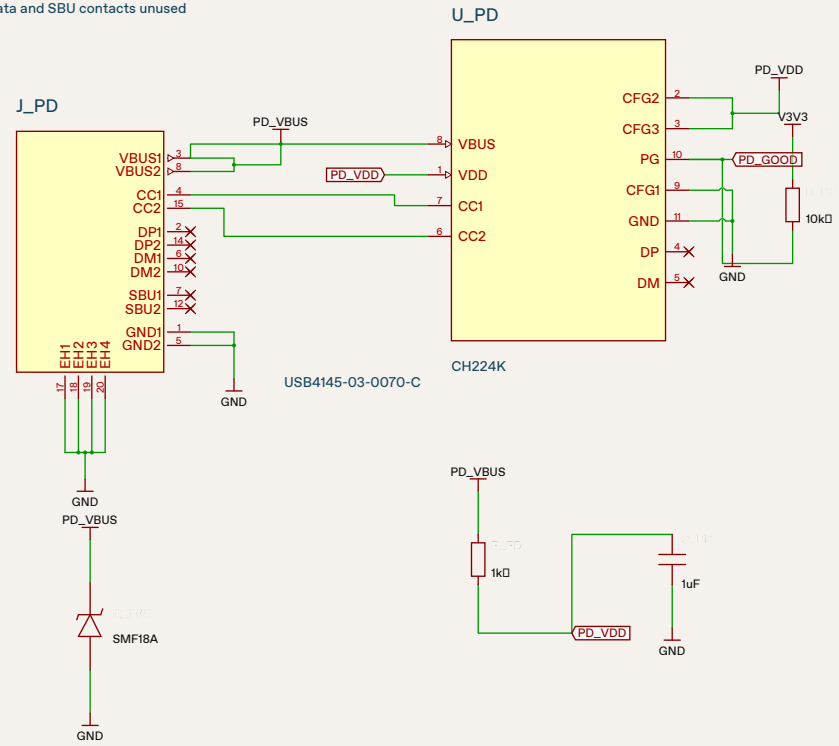
NEMA14 / RP2040

Power & USB-PD

Motor supply negotiation, reverse-blocked logic power and 3.3 V conversion

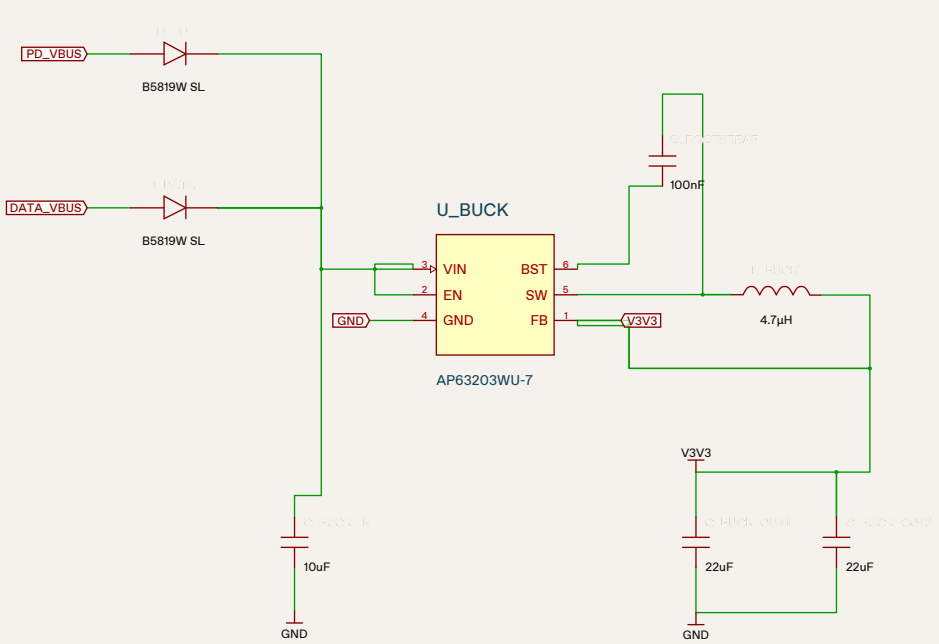
USB-C motor power

Fixed 15 V request; data and SBU contacts unused



3.3 V logic supply

AP63203 fixed-output buck; USB power is diode-isolated



PD_VBUS = motor rail DATA_VBUS = programming USB 5 V

VIN range and transients require bench qualification.

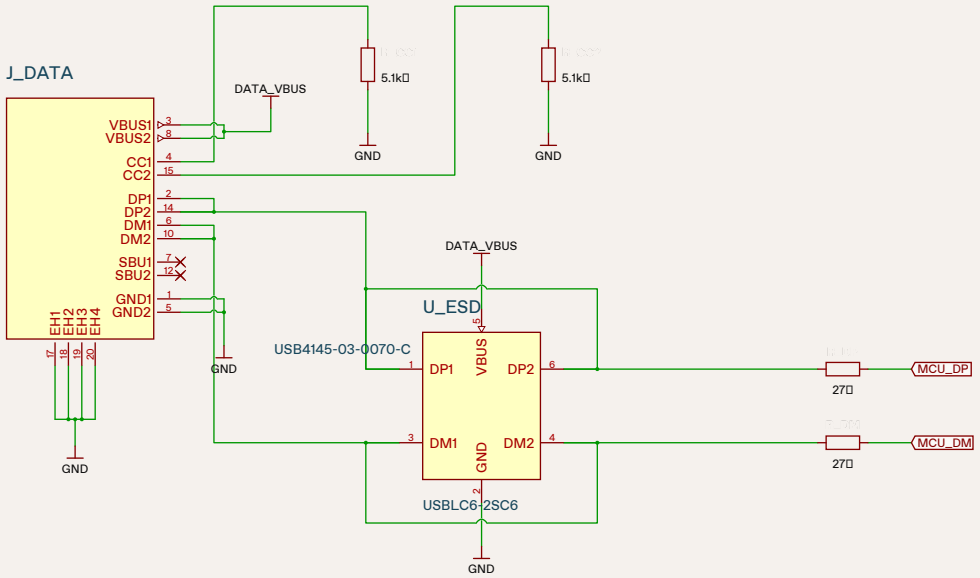
NEMA14 / RP2040

Programming & USB

Programming USB-C, data-line protection and nonvolatile boot interfaces

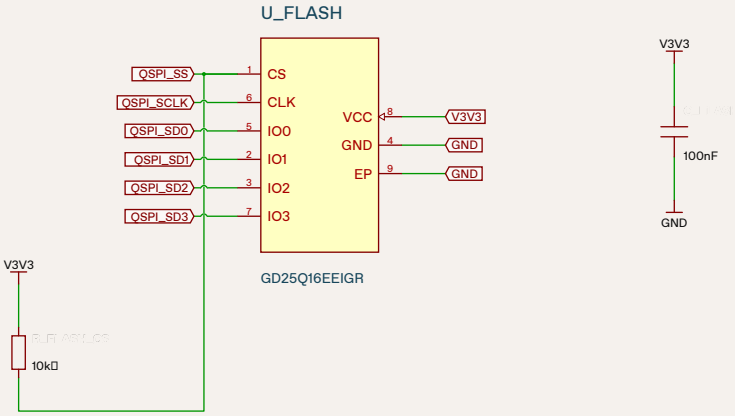
USB-C data port

Separate 5.1 kΩ CC pull-downs; duplicated D+/D− contacts



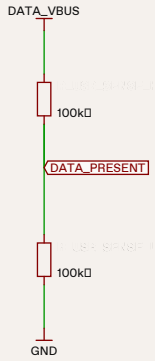
QSPI flash & boot selection

GD25Q16E; BOOT contact asserts the boot input



USB presence

Resistive divider to MCU; fixed 5 V input only



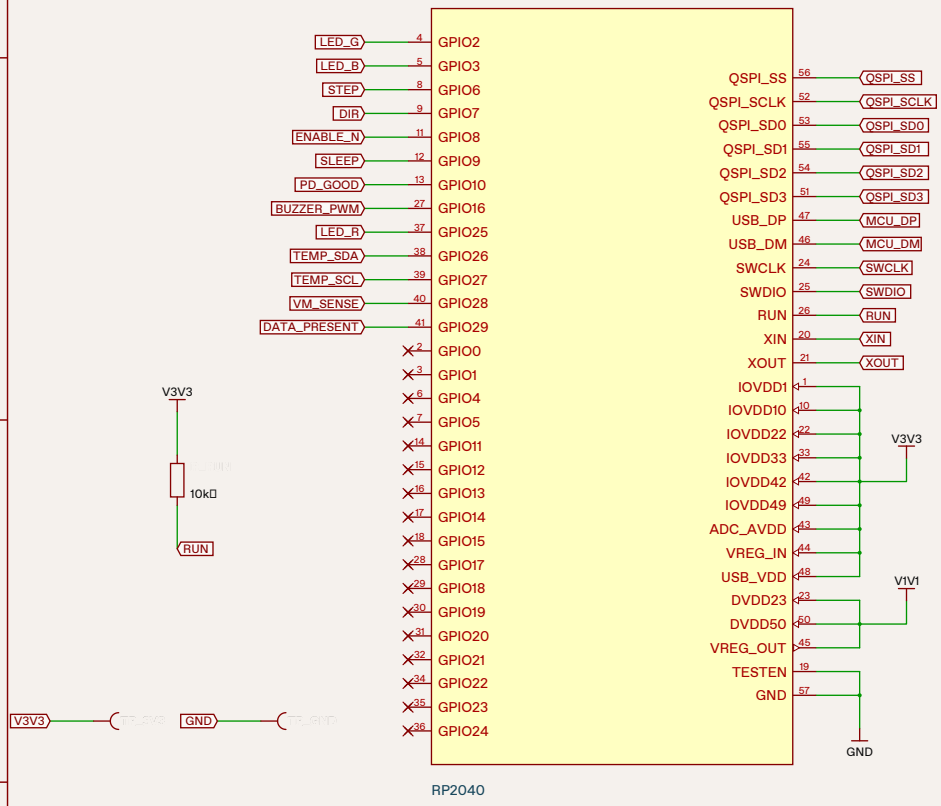


NEMA14 / RP2040

Controller & clocks

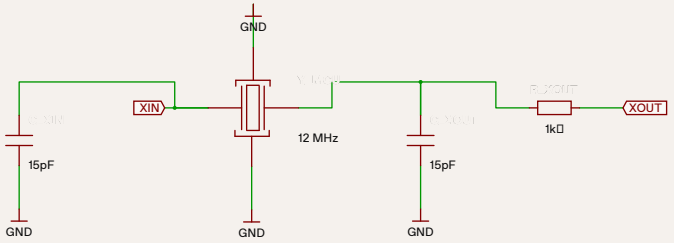
MCU interfaces, reset/debug access, power decoupling and clock support

U_MCU



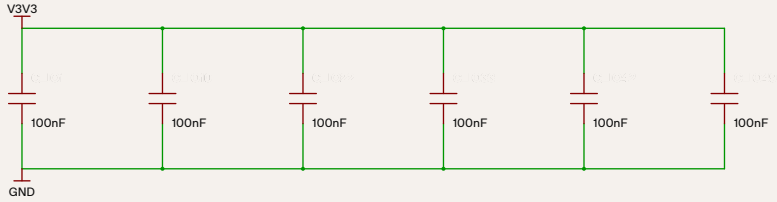
12 MHz crystal

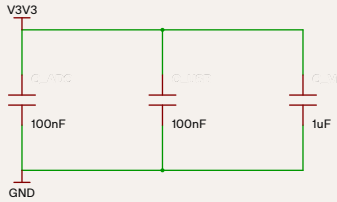
Native 4-pin crystal; load capacitors and series resistor



3.3 V bypass

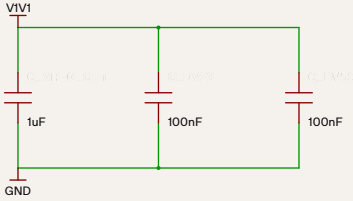
IOVDD, ADC and USB decoupling





1.1 V core

On-chip VREG_OUT feeds DVDD



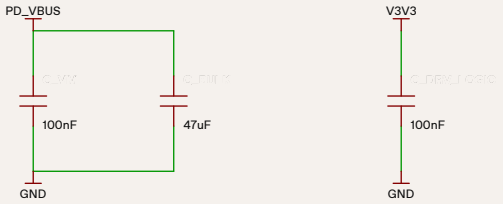
NEMA14 / RP2040

Stepper motor driver

A4988 phase outputs, charge pump, current limit and motor-voltage measurement

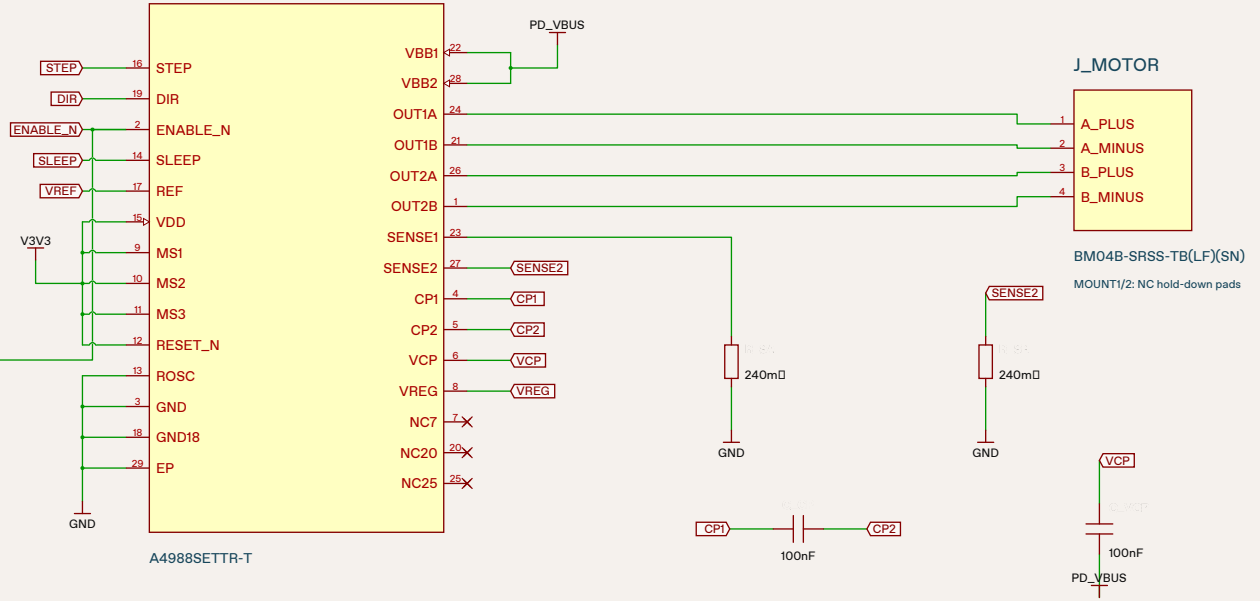
Driver supplies

Local bulk and high-frequency bypass



JST-SH pin order: 1 A+ 2 A– 3 B+ 4 B–

U_DRV



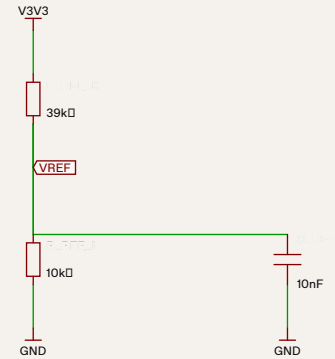
J_MOTOR

BM04B-SRSS-TB(LF)(SN)

MOUNT1/2: NC hold-down pads

Current reference

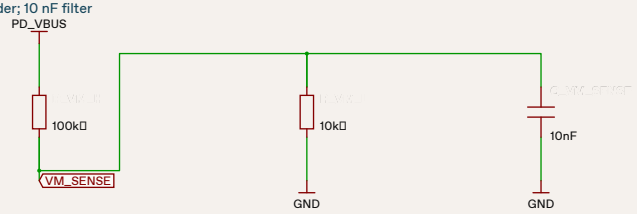
VREF / (8 × RSENSE) = 0.351 A nominal





Motor voltage monitor

100 kΩ / 10 kΩ divider; 10 nF filter





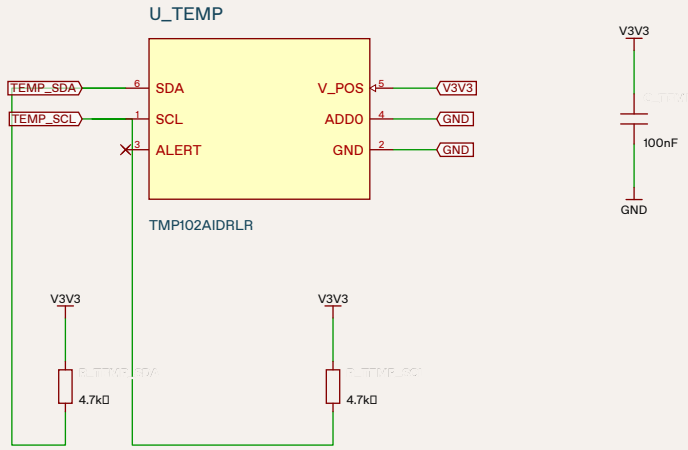
NEMA14 / RP2040

Temperature & feedback

I²C temperature telemetry, RGB status outputs and PWM magnetic buzzer

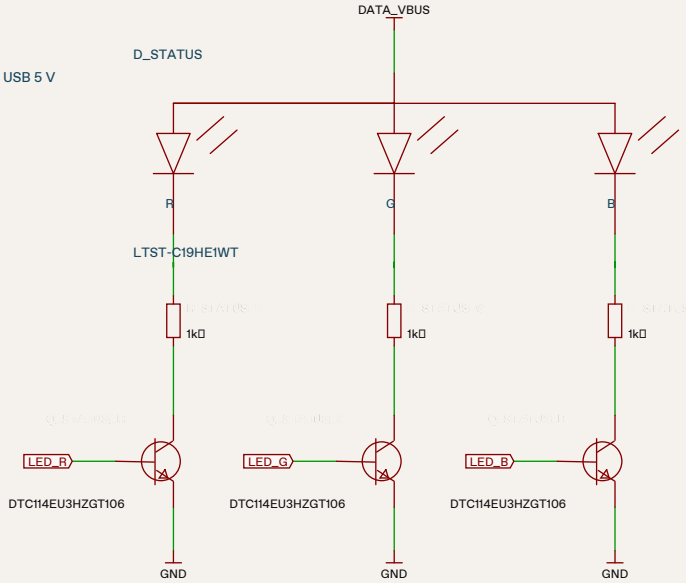
TMP102 temperature

ADD0 grounded: address 0x48; ALERT unused



RGB status indicator

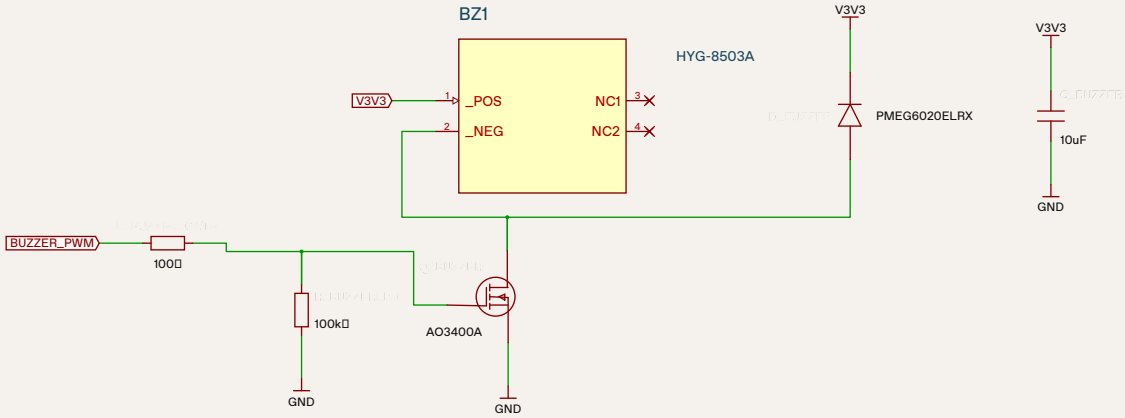
Common anode uses programming USB 5 V



DTC114E: NPN with internal input/bias resistors.

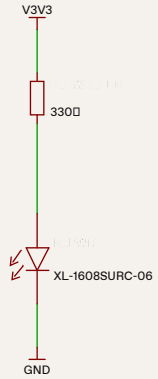
PWM buzzer

120 mA magnetic load; MOSFET drive and flyback diode



Power indication

3.3 V rail present; RGB/buzzer require firmware



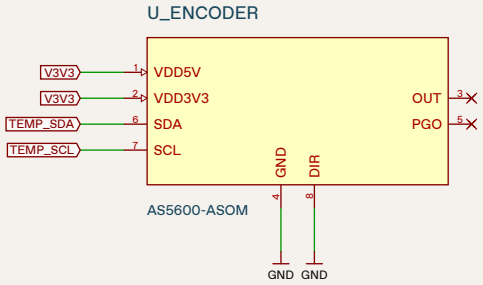
NEMA14 / RP2040

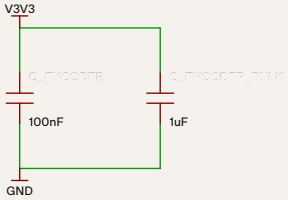
Optional encoder (DNP)

Bottom footprints only; AS5600 and both bypass capacitors omitted from factory assembly

Future magnetic shaft position

Nominal sensor centre (0, 0); rear shaft/magnet geometry remains unverified





U_ENCODER, C_ENCODER and C_ENCODER_BULK: bottom / DO NOT PLACE.

3.3 V mode: VDD5V + VDD3V3 tied together; DIR grounded; OUT/PGO NC.

Shared existing temperature I²C bus: AS5600 0x36 / TMP102 0x48.

Existing 4.7 k□ top-side pull-ups retained. Encoder firmware is required.

A diametrically magnetized shaft magnet and motor fit must be validated.